Can a Flame Persist After Its Light Has Gone?

A Quantum-Kinetically Informed Hypothesis for Cool-Flame Persistence in Microgravity

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Research hypothesis and technical framework

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Abstract

Microgravity droplet-combustion experiments have revealed a counterintuitive state: after the visible hot flame undergoes radiative extinction, some large *n*-alkane droplets continue to vaporize quasi-steadily before a later, distinct extinction. This post-luminous regime is commonly interpreted through low-temperature, negative-temperature-coefficient (NTC) chemistry. We propose a testable cross-scale hypothesis: quantum nuclear tunnelling in selected hydrogen-transfer steps of low-temperature oxidation may measurably shift the persistence and extinction boundaries of this hidden reaction regime, while microgravity modifies the transport environment that makes the regime observable. The claim is not that a macroscopic flame is a quantum object, nor that gravity directly enhances tunnelling. Rather, molecular barrier-crossing corrections may alter elementary rate constants which, when coupled to diffusion, radiation and droplet heat transfer, change classical continuum-scale outcomes. We formulate a reaction–transport model, identify the FLEX data as the primary retrospective test bed, and distinguish the roles of FlameLens dataset families for extinction, flame spread and smoke-related phenomena. Existing public records can constrain macroscopic signatures but do not directly measure tunnelling or all relevant radical intermediates. We therefore specify discriminating predictions, isotope-substitution experiments, model-comparison procedures and explicit falsification criteria. The central open question is whether a quantum-corrected description of selected elementary reactions can explain otherwise unresolved variation in post-luminous combustion persistence without sacrificing the predictive accuracy of classical transport models. **Keywords:** microgravity combustion; cool flames; quantum tunnelling; hydrogen transfer; negative-temperature-coefficient chemistry; droplet extinction; kinetic isotope effect; multiscale modelling

1 INTRODUCTION

Combustion in microgravity provides an unusual opportunity to separate chemical kinetics from buoyancy-dominated transport. In normal gravity, thermal expansion generates buoyant flow that can distort flame geometry, alter residence time and obscure transport–chemistry coupling. Suppressing buoyancy does not remove transport; it changes its relative composition, leaving diffusion, imposed flow, Stefan flow, radiation and residual accelerations to determine the local environment. NASA's combustion research guide identifies this simplification as a central reason for conducting combustion experiments in orbit [4].

The Flame Extinguishment Experiment (FLEX) aboard the International Space Station (ISS) established an especially intriguing observation. For sufficiently large *n*-alkane droplets, the visible hot flame may disappear by radiative extinction while the droplet continues to vaporize for an extended period. The second stage eventually ends at a finite droplet diameter, accompanied by formation of a surrounding vapour cloud [2, 3, 1]. This behaviour was interpreted as cool-flame-supported, low-temperature burning associated with NTC chemistry. It is a crucial reminder that visible luminosity, heat release and chemical conversion are not interchangeable observables.

The proposed research asks whether that post-luminous interval contains a molecular-scale sensitivity that is not adequately represented by conventional classical rate expressions. Quantum tunnelling is a well-established correction for selected reactions involving light atoms, especially hydrogen, and can become important when thermal activation is insufficient to describe barrier crossing [5]. Low-temperature hydrocarbon oxidation also contains internal hydrogen-transfer steps involving alkylperoxy and hydroperoxyalkyl radicals, RO_2 and QOOH, that control branching and subsequent oxidation [6, 7].

The proposed connection is deliberately narrow. The question is not whether a flame exhibits macroscopic quantum coherence, nor whether microgravity changes fundamental quantum mechanics. The question is whether quantum corrections to a small set of elementary hydrogen-transfer rate constants can produce a measurable difference in the persistence of a macroscopically classical, transport-coupled cool-flame regime.

1.1 Research question

> *Can quantum-tunnelling-sensitive hydrogen-transfer kinetics shift the boundary between visible-flame extinction, post-luminous cool-flame persistence and complete chemical extinction in microgravity droplet combustion?*

The question is open-ended by design. The proposed mecha-

nism may be important, negligible, or masked by transport and measurement uncertainties. All three outcomes are scientifically useful.

1.2 Scope within FlameLens

FlameLens brings together several kinds of combustion evidence. The hypothesis is anchored in FLEX droplet extinction because it contains the clearest published evidence of a sustained low-temperature regime after visible flame extinction. Other project families have narrower roles: CFI-G provides gas-phase normal and inverse cool-diffusion-flame cases; BASS-II and the SAFFIRE series inform solid-fuel flame growth, spread and extinction; SPICE informs non-buoyant jet-flame smoke-point and sooting behaviour; and SAME concerns smoke-aerosol properties and transport. These are complementary fire-safety phenomena, not interchangeable measurements of the same reaction pathway [9, 10, 12, 11, 13, 14].

2 PHYSICAL BASIS AND CENTRAL HYPOTHESIS

2.1 The post-luminous combustion interval

For a droplet of instantaneous diameter $D(t)$, quasi-steady droplet combustion is often described over a limited interval by the D^2 law,

$$D^2(t) = D_0^2 - Kt, \tag{1}$$

where D_0 is a reference diameter and K is an effective regression constant. In the FLEX cool-flame observations, continued approximately quasi-steady regression after visible radiative extinction was used as evidence of a second combustion stage [2, 3]. Equation (1) is a useful macroscopic descriptor, but it is not by itself proof of a particular radical sequence or a direct measurement of the chemical heat-release rate.

We distinguish three event times:

$$t_L : \text{loss of the visible luminous hot flame,} \tag{2}$$

$$t_C : \text{cessation of measurable chemical conversion,} \tag{3}$$

$$t_V : \text{cessation of measurable droplet regression or vaporization.} \tag{4}$$

In an ideal diagnostic configuration, t_C and t_V would be independently estimated rather than assumed equal. The post-luminous persistence interval is defined as

$$\Delta t_{PL} = t_C - t_L, \tag{5}$$

provided a nonzero interval is supported by species-sensitive or calorimetric evidence. If only optical data and droplet size are available, Δt_{PL} must be reported as an inferred interval with a corresponding uncertainty, not as a directly observed chemical lifetime.

2.2 Low-temperature oxidation and hydrogen transfer

A simplified low-temperature alkane oxidation sequence is

$$\mathrm{RH + X \rightarrow R + XH}, \tag{6}$$

$$\mathrm{R + O_2 \rightleftharpoons RO_2}, \tag{7}$$

$$\mathrm{RO_2 \rightarrow QOOH}, \tag{8}$$

$$\mathrm{QOOH + O_2 \rightarrow O_2QOOH}, \tag{9}$$

$$\mathrm{O_2QOOH \rightarrow KHP + OH + \cdots}. \tag{10}$$

Here RH denotes fuel, R an alkyl radical, RO_2 an alkylperoxy radical, QOOH a hydroperoxyalkyl radical and KHP a ketohydroperoxide class of intermediate. This scheme is schematic: a detailed mechanism contains multiple isomers, competing channels, pressure-dependent association and decomposition, and fuel-specific pathways. The formation and reaction of RO_2, QOOH and O_2QOOH are central to low-temperature hydrocarbon oxidation [6, 7].

The intramolecular H-shift in Eq. (8) and selected H-abstraction steps are candidates for quantum nuclear effects. Their importance cannot be inferred from the existence of a cool flame alone. It must be computed for the relevant molecular structure and tested against observables.

2.3 Quantum tunnelling as a rate correction

Conventional transition-state theory gives a thermal rate estimate $k_j^{\mathrm{TST}}(T)$. A quantum-corrected rate may be written

$$k_j^{\mathrm{Q}}(T) = \kappa_j(T)\,\Gamma_j^{\mathrm{CVT}}(T)\,k_j^{\mathrm{TST}}(T), \tag{11}$$

where κ_j is a tunnelling transmission coefficient and Γ_j^{CVT} represents a variational recrossing correction. The exact factorization and computational method depend on the chosen kinetics framework. Small-curvature tunnelling (SCT), multidimensional semiclassical methods or instanton approaches are preferable when a simple parabolic-barrier approximation is inadequate. Recent combustion-kinetics work illustrates the use of variational transition-state theory and SCT corrections for hydrogen abstraction from fuel molecules, including alkanes; it does not, however, establish the correction for the specific FLEX radical pathways [8].

For a first-order estimate only, the Wigner correction is

$$\kappa_W(T) = 1 + \frac{1}{24}\left(\frac{\hbar|\omega^{\ddagger}|}{k_B T}\right)^2, \tag{12}$$

where $|\omega^{\ddagger}|$ is the magnitude of the imaginary angular frequency at the transition state, $\hbar$ is the reduced Planck constant and k_B is Boltzmann's constant. Equation (12) is a local harmonic approximation, not a validated rate expression for the FLEX chemistry. It is included to make the temperature and barrier-frequency dependence explicit; a publication-quality mechanism must compute pathway-specific corrections.

Hydrogen/deuterium substitution provides a discriminating diagnostic. The kinetic isotope effect is

$$\mathrm{KIE}(T) = \frac{k_H(T)}{k_D(T)}, \tag{13}$$

where k_H and k_D refer to matched isotopologues and the same elementary channel. A measured isotope effect is not automatically proof of tunnelling: zero-point energy, conformational changes, diffusion and other isotope-sensitive effects must be included in the interpretation. However, a pre-registered isotope-dependent prediction from a quantum-corrected mechanism provides a much stronger test than fitting unlabelled droplet histories alone.

2.4 What microgravity changes—and what it does not

Microgravity changes the transport problem by suppressing buoyancy-driven flow. It does not directly reduce an activation barrier or switch on tunnelling. A schematic species equation is

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla\cdot(\rho\mathbf{u}Y_i) = \nabla\cdot(\rho D_i\nabla Y_i) + \dot{\omega}_i, \tag{14}$$

where Y_i is species mass fraction, D_i its effective diffusivity, ρ density, $\mathbf{u}$ velocity and $\dot{\omega}_i$ the chemical source term. The source term depends on the rate constants in Eqs. (11)–(10); the transport operator depends on flow, diffusion and geometry.

A corresponding energy equation is

$$\rho c_p\frac{DT}{Dt} = \nabla\cdot(\lambda\nabla T) - \nabla\cdot\mathbf{q}_r - \sum_i h_i\dot{\omega}_i, \tag{15}$$

where c_p is heat capacity, λ thermal conductivity, $\mathbf{q}_r$ radiative heat flux and h_i species enthalpy (sign convention absorbed into the source term). Microgravity changes the velocity field and residence-time distribution, while the quantum correction changes selected chemical source terms. Their coupling is the object of study.

2.5 Central hypothesis

H1 (primary hypothesis). In the low-temperature post-luminous regime of selected *n*-alkane droplets, quantum tunnelling corrections to one or more hydrogen-transfer reactions in the low-temperature oxidation network produce a measurable change in predicted cool-flame persistence and/or the final extinction condition, relative to a consistently calibrated classical-rate model, after transport and radiative losses are constrained.

H2 (isotope prediction). At matched macroscopic conditions, site-specific deuteration of a hydrogen-transfer site identified by sensitivity analysis produces an isotope-dependent change in post-luminous persistence or extinction behaviour whose direction and magnitude are predicted in advance by the quantum-kinetic mechanism.

H3 (microgravity observability). Suppression of buoyancy-driven transport improves the ability to isolate the coupling between low-temperature chemistry and diffusion/radiation. This is an observability hypothesis, not a claim that microgravity increases the intrinsic tunnelling probability.

3 A MULTISCALE MODEL FOR TESTING THE HYPOTHESIS

3.1 Reaction–transport model hierarchy

The minimum useful model should contain: (i) transient spherical droplet mass and energy balances; (ii) gas-phase species and energy conservation; (iii) a validated high- and low-temperature fuel mechanism; (iv) radiation and liquid-phase transport; and (v) alternative classical and quantum-corrected elementary rate sets. For a spherical gas-phase control volume, Eq. (14) may be solved with radial symmetry only when the observed flow and apparatus permit that approximation. Residual acceleration, droplet drift, tether effects and imposed flow must be checked before imposing spherical symmetry.

For an elementary pathway j, define a correction ratio

$$\chi_j(T) = \frac{k_j^{\mathrm{Q}}(T)}{k_j^{\mathrm{cl}}(T)}. \tag{16}$$

The mechanism-level sensitivity is not determined by χ_j alone. It also depends on the pathway's flux, branching competition, reverse reactions and coupling to heat release. A pathway with a large tunnelling correction may have negligible macroscopic influence if its reaction flux is small.

3.2 Timescale and transport diagnostics

A useful first-order ratio is the Damköhler number

$$\mathrm{Da} = \frac{\tau_{\mathrm{res}}}{\tau_{\mathrm{chem}}}, \tag{17}$$

where τ_{res} is a characteristic residence or transport time and τ_{chem} a chemical timescale defined for the selected low-temperature pathway. The definition must be fixed before comparing experiments. In the absence of strong buoyancy, diffusion and any forced flow can dominate transport. The Grashof number, $\mathrm{Gr} = g\beta\Delta TL^3/\nu^2$, provides a scale for buoyancy relative to viscous effects; reducing effective g reduces this contribution, but does not guarantee zero residual acceleration or eliminate other flow mechanisms.

3.3 State labels and observability

A modelling system must not use a single binary label called "extinguished" for all outcomes. At minimum, the state space should distinguish:

1. visible hot-flame persistence;
2. visible radiative extinction;
3. post-luminous chemical or cool-flame persistence;
4. complete chemical extinction;
5. disruptive extinction or droplet disruption; and
6. unresolved/measurement-censored termination.

Camera radiance, broadband radiometry, droplet diameter history and species diagnostics should remain separate observations with separate uncertainty models. Continued droplet regression alone is an indirect proxy; it cannot, on its own, identify the radical sequence.

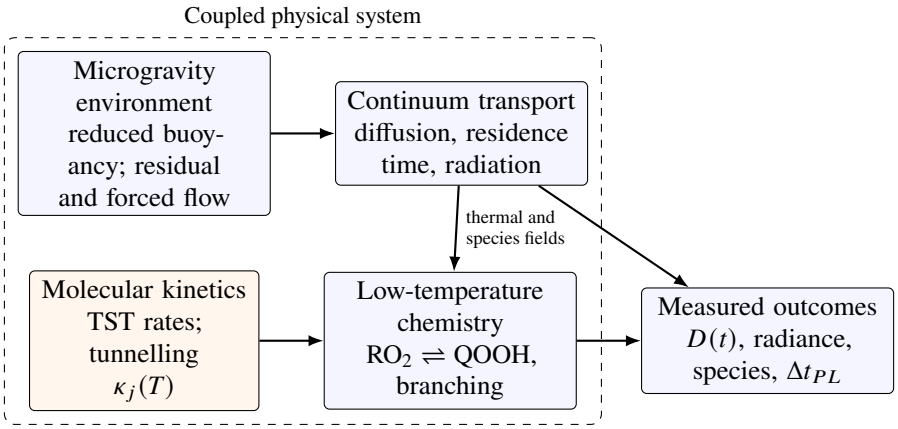


Figure 1: Proposed cross-scale coupling. Microgravity affects transport, while quantum nuclear effects enter only through selected elementary reaction rates. The observable flame state remains a classical continuum-scale outcome. Diagram is a conceptual model, not a measured causal result.

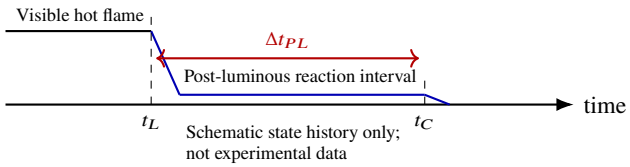


Figure 2: Event distinction required by the hypothesis. The loss of visible luminosity at t_L is not assumed to coincide with complete chemical extinction at t_C. The post-luminous interval must be established with independent diagnostics.

4 DATA STRATEGY WITHIN FLAMELENS

4.1 Primary retrospective dataset: PSI-69 FLEX

NASA's PSI-69 FLEX Version 5 record identifies the investigation as an ISS microgravity droplet-combustion study and assigns DOI `10.60555/mbq8-0451`. The repository record is marked CC0-1.0 and displays a 275-entry experimental summary table with fuel, pressure, initial oxygen/inert-gas composition, initial droplet diameter, visible extinction diameter, burning rate, burn time and test-end outcome among its fields [9]. Published FLEX summaries report tests with heptane and methanol and describe cool-flame-supported second-stage burning for large *n*-alkane droplets after visible hot-flame extinction [1, 2, 3].

The table-level metadata are suitable for an initial audit, condition-matched comparisons and event-label harmonisation. They are not sufficient to estimate elementary tunnelling coefficients. The archived image, radiometer, calibration and raw data products must be located and inspected before making claims about time-resolved chemical states. The PSI page also lists raw-data and science-document collections; their contents and version history must be frozen and cited before a reproducible analysis begins [9].

4.2 Other FlameLens evidence families

Table 1 defines the correct role of the wider FlameLens evidence portfolio. These datasets can test whether the transport and event-interpretation framework generalises across combustion configurations. They do not all contain the $RO_2 \rightarrow QOOH$ chemistry central to the proposed hypothesis.

4.3 Data audit and model comparison

The retrospective study should proceed in the following order:

1. Freeze source versions, file hashes, metadata, units, calibration documents and test identifiers. Preserve the NASA persistent identifier for each investigation.
2. Separate fuel families. Do not pool methanol with *n*-alkanes for the cool-flame hypothesis merely because both appear in FLEX. The published cool-flame observation is specifically associated with large *n*-alkane droplets.
3. Define outcomes at the experiment level. Preserve visible radiative extinction, diffusive extinction, continued second-stage regression, disruption and censoring as distinct labels.
4. Harmonise pressure, oxygen fraction, diluent identity, initial diameter, fuel, deployment type and other available control variables. Record missingness rather than silently imputing scientific event labels.
5. Compare a classical mechanism (M0) and a quantum-corrected mechanism (M1) using identical transport assumptions, parameter priors and calibration budgets. Only the selected elementary rate treatment should differ.
6. Split and validate by independent experiment/run groups, not randomly by correlated image frames or time samples. Perform leave-one-condition-family-out checks where data coverage permits.

The principal model-comparison quantities should include held-out predictive log score, calibration of event probabilities, error in $D(t)$ and extinction diameter, and uncertainty in Δt_{PL}. A more complex model is not preferred simply because it fits the calibration set better. It must improve prediction on genuinely held-out experiments and survive sensitivity analysis.

Table 1: Role of FlameLens evidence families in the proposed research. Dataset identifiers and scientific targets are separated to avoid treating unlike endpoints as equivalent.

Evidence family	Representative PSI investigation	Appropriate role	What it cannot establish by itself
Droplet extinction and cool-flame persistence	PSI-69 FLEX	Primary retrospective test of visible extinction, continued droplet regression and later extinction in microgravity	Direct tunnelling probability, radical concentrations or isotope effects
Gas-phase cool diffusion flames	PSI-159 ACME CFI-G (normal and inverse cases kept distinct)	Secondary cool-flame comparison across spherical diffusion-flame configurations and flow orientations	Equivalence to isolated-droplet FLEX chemistry or direct proof of tunnelling
Solid-fuel flame spread	PSI-25 BASS-II; SAFFIRE series	Transport-level stress test for ignition, flame growth, spread and extinction on solid materials	The alkane droplet RO_2/QOOH mechanism
Soot point and sooting propensity	PSI-107 SPICE	Separate study of residence-time, flow and fuel effects on luminous flame and smoke-point outcomes	Cool-flame chemical persistence or tunnelling in the FLEX pathways
Smoke aerosol and detection context	PSI-102 SAME	Separate analysis of smoke particle-size and transport measurements relevant to spacecraft fire detection	Soot production rate, hot-flame extinction or elementary reaction tunnelling

5 FALSIFIABLE PREDICTIONS

The hypothesis is useful only if it can fail. The following predictions should be registered before model fitting or isotope experiments.

P1: pathway-specific kinetic sensitivity. Quantum-chemical calculations should identify at least one relevant H-transfer pathway for which $\kappa_j(T)$ differs from unity by more than the propagated uncertainty over the temperature range actually visited by the post-luminous regime. If all plausible pathway corrections are negligible over that range, the proposed mechanism loses its primary physical basis.

P2: macroscopic predictive value. Holding the transport model fixed, M1 should improve held-out predictions of one or more of the following: post-luminous regression rate, final cool-flame extinction diameter, probability of second-stage persistence, or its duration. The improvement must exceed uncertainty from fuel properties, radiation, initial droplet size, residual acceleration and diagnostic timing.

P3: isotope sensitivity. A matched H/D experiment should show an isotope-dependent response in a target observable. The sign and size of the response must be predicted by the chosen reaction mechanism before inspecting the isotope data. A post-hoc isotope narrative is not a valid test.

P4: transport–chemistry interaction. The effect of M1 should be strongest in the regime where low-temperature chemical timescales and transport/radiative timescales are competitive. A quantum correction that improves only one unreplicated corner of parameter space is insufficient evidence for a general mechanism.

P5: cross-configuration restraint. Transport-level patterns may recur in BASS-II or SPICE, but the molecular tunnelling correction derived for an alkane droplet mechanism should not be transferred unchanged to solid-fuel spread or jet diffusion flames. Cross-family transfer requires a new mechanism and new evidence.

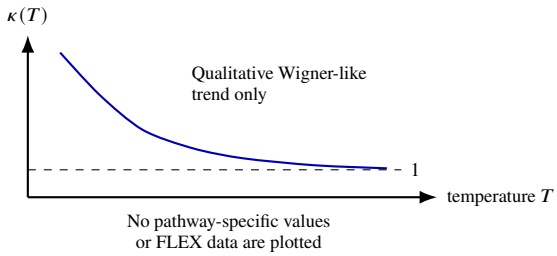


Figure 3: Illustrative temperature dependence of a tunnelling transmission coefficient. The curve is schematic and is not fitted to an alkane reaction or experimental data. Actual $\kappa_j(T)$ must be calculated for each elementary pathway.

6 PROPOSED EXPERIMENTAL PROGRAM

6.1 Stage I: retrospective FLEX analysis

Reconstruct the second-stage interval from available time-resolved droplet images, radiometry and supporting diagnostics. Independently identify visible hot-flame extinction and the end of measurable reaction. Where direct chemical diagnostics are absent, label the endpoint as an inferred proxy. Use the published heptane, octane and decane observations as motivation, but do not assume identical kinetic mechanisms across fuels [3].

6.2 Stage II: quantum-kinetic mechanism development

For candidate elementary pathways, perform conformer searches, electronic-structure calculations, transition-state verification, intrinsic reaction-coordinate checks, and temperature- and pressure-dependent rate calculations. Compare conventional TST, variational TST with SCT, and another validated tunnelling treatment over the experimentally relevant temperature

range. Include uncertainty from conformers, barrier heights, vibrational frequencies, pressure dependence and competing channels. A rate correction should not be added as a global empirical multiplier to the entire combustion mechanism.

6.3 Stage III: isotope-resolved microgravity experiment

A decisive experiment would compare an ordinary fuel with a site-specifically deuterated analogue selected from the pathway-sensitivity analysis. The two fuels should be tested under matched initial droplet diameter, pressure, oxygen fraction, diluent composition, ignition protocol and apparatus configuration. Replicates should be randomised across test order where feasible. A normal-gravity or controlled-flow reference may help isolate transport changes, but it cannot replace the microgravity comparison.

Required measurements include high-dynamic-range imaging, droplet diameter histories, broadband and preferably spectrally resolved radiometry, local temperature constraints, and species-sensitive diagnostics chosen to observe selected low-temperature intermediates or products. Candidate species include formaldehyde, hydroxyl, hydroperoxy and peroxide-related products; diagnostic feasibility and spectral interference must be established before payload design. A detector that only observes visible luminosity cannot establish complete chemical extinction.

6.4 Stage IV: FlameLens analysis integration

FlameLens should represent this hypothesis as a traceable model comparison rather than a single black-box score. Each result should link to its source test, dataset version, units, fuel identity, event definition, kinetic mechanism version, parameter set, code revision and uncertainty components. Model outputs should display classical and quantum-corrected predictions side by side, with separate uncertainty bands for measurement, dataset coverage, kinetics and model structure.

A model trained on FLEX outcomes may estimate the probability of a post-luminous regime within its demonstrated domain. It must not label such a probability as direct evidence of tunnelling. SHAP or other feature-attribution methods can describe associations in the fitted predictor, but cannot establish that a quantum mechanism caused the observed outcome.

7 DATA LIMITATIONS AND IDENTIFIABILITY

The central identifiability problem is that a macroscopic droplet history is the result of coupled chemistry, transport, radiation and liquid-phase processes. Several different mechanisms can produce similar $D(t)$ curves. A retrospective dataset containing fuel, pressure, oxygen fraction, initial size, visible extinction diameter and burn time can constrain the phenomenology, but it cannot uniquely identify a tunnelling transmission coefficient. The primary PSI-69 metadata record itself separates experimental tables from raw data, calibration and science-document collections; these must be audited individually before analysis [9].

The following uncertainties must remain separate:

- **Measurement uncertainty:** image segmentation, radiometer calibration, timing and sensor detection limits.
- **Coverage uncertainty:** missing fuels, pressures, diluents, droplet sizes, incomplete histories and unbalanced condition families.
- **Kinetic uncertainty:** elementary rate constants, conformers, barrier heights and pressure-dependent branching.
- **Transport and radiation uncertainty:** residual acceleration, droplet drift, Stefan flow, diffusion, heat losses and geometry.
- **Structural uncertainty:** omitted reactions, simplified symmetry assumptions and imperfect coupling between liquid and gas phases.

There is also a transferability limit. FLEX cool-flame persistence is most directly relevant to large alkane droplets. Methanol has different chemistry and, in the cited FLEX overview, did not exhibit the same cool-flame behaviour as large heptane droplets under the reported conditions [1]. A quantum correction derived for an alkylperoxy H-shift must not be assumed to apply to soot inception, smoke aerosol growth, solid-fuel flame spread or smoke-detector response.

8 DISCUSSION: WHAT WOULD MAKE THIS IMPORTANT?

If supported, the result would not mean that a spacecraft flame is a quantum state. It would show that a molecular quantum effect in a small subset of elementary reactions can influence a macroscopic fire-safety observable after transport and heat-loss processes are coupled. This would connect quantum chemistry, low-temperature oxidation, microgravity transport and fire-state interpretation in a quantitatively testable way.

The most important possible outcome may be negative. If a classical mechanism, after careful calibration and uncertainty propagation, predicts post-luminous persistence as well as or better than the quantum-corrected mechanism, then tunnelling is not a necessary explanation for the observed regime at those conditions. The result would constrain where quantum corrections matter and prevent an attractive but unsupported quantum narrative from entering spacecraft fire models.

A positive result also requires restraint. Improved prediction alone does not prove that tunnelling is the unique explanation; it must be accompanied by pathway-specific quantum calculations and preferably an isotope-sensitive experiment. Conversely, a measured isotope effect alone is insufficient unless transport, zero-point energy, thermochemistry and fuel-property differences are controlled. The combination of mechanism calculation, macroscopic prediction and independent isotope response is the decisive evidence chain.

9 CONCLUSIONS

This paper proposes an open, testable hypothesis: quantum tunnelling in selected hydrogen-transfer reactions may shift the

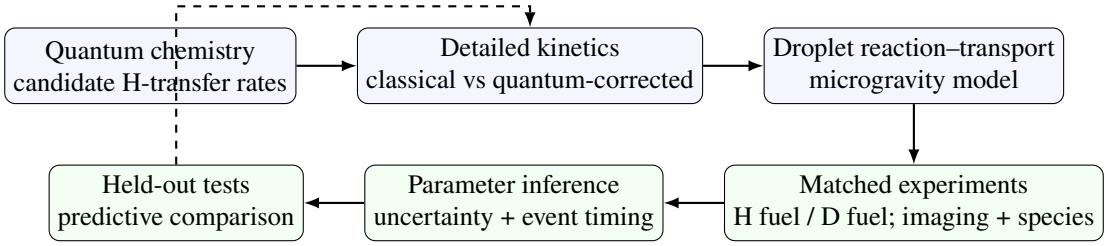


Figure 4: Proposed validation loop. Molecular calculations supply pathway-specific rate corrections; detailed kinetics and transport predict observables; matched experiments test those predictions; held-out validation decides whether the quantum correction adds explanatory value.

persistence and extinction behaviour of low-temperature cool flames in microgravity droplet combustion. The proposal is motivated by the FLEX observation that some large alkane droplets continue to burn or vaporize after visible hot-flame extinction, and by the established role of low-temperature RO_2/QOOH chemistry in hydrocarbon oxidation. Microgravity is treated as a transport-control environment, not as a source of enhanced quantum tunnelling.

The hypothesis requires more than re-fitting existing flame histories. It needs pathway-specific quantum kinetics, a transient reaction–transport model, clear separation of visible and chemical extinction, matched isotope experiments, and run-grouped out-of-sample validation. FLEX provides the primary retrospective anchor; FlameLens's flame-spread, soot and smoke families provide distinct context and transport-level tests rather than direct evidence of the proposed molecular mechanism.

The question remains deliberately unresolved: *when the visible flame disappears, could a quantum-sensitive molecular reaction network be helping the combustion process persist in a form that conventional optical fire interpretation fails to recognise?* A well-designed negative experiment would be as informative as a positive one.

DATA AND CODE AVAILABILITY

This manuscript is a hypothesis and methods proposal. No new experiments, fitted kinetic parameters, model scores or original numerical results are claimed. The principal retrospective source is NASA PSI-69 FLEX (DOI: `10.60555/mbq8-0451`); additional public context includes PSI-159 ACME CFI-G, PSI-98 SAFFIRE-I (and related PSI-99/PSI-100 investigations), PSI-25 BASS-II, PSI-107 SPICE and PSI-102 SAME. NASA PSI requests citation of each investigation's persistent identifier. The live FlameLens URL provided for this draft resolves to a sign-in route; protected dashboard contents and deployed model outputs were therefore not independently inspected. The current draft does not claim that any private dashboard result has been reproduced.

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